

```
1 import numpy as np
2
3 def rotate_3d_coordinates(coordinates, x_degrees,
4 y_degrees, z_degrees):
5     """
6     Rotates a set of 3D coordinates around the x, y,
7     and z axes by specified angles.
8
9     Args:
10     coordinates (np.array): An N x 3 array of 3D
11     coordinates.
12     x_degrees (float): The rotation angle around the
13     x-axis in degrees.
14     y_degrees (float): The rotation angle around the
15     y-axis in degrees.
16     z_degrees (float): The rotation angle around the
17     z-axis in degrees.
18
19     Returns:
20     np.array: The rotated coordinates: An N x 3 array
21     of 3D coordinates.
22     """
23     #TODO: write this function
24     Rx = np.array([[1, 0, 0],[0, np.cos(x_degrees), -
25 1 * np.sin(x_degrees)],[0, np.sin(x_degrees), np.cos(
26 x_degrees)]]))
27     Ry = np.array([[np.cos(y_degrees), 0, np.sin(
28 y_degrees)],[0,1, 0],[-1* np.sin(y_degrees), 0, np.
29 cos(y_degrees)]]))
30     Rz = np.array([[np.cos(z_degrees), -1 * np.sin(
31 z_degrees),0],[ np.sin(z_degrees), np.cos(z_degrees),
32 0],[0, 0, 1]])
33     rotated_coordinates = coordinates @ (Rx @ Ry @ Rz
34 ).T
35
36     return rotated_coordinates
37
```